Template Document

Academic Support Agent

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Abstract

Your abstract must be placed between the beginabstract and endabstract tags. The abstract is a 200 - 300 word concise and clear summary of:

1. What you set out to do and why,
2. How you did it,
3. What you found,
4. (Optionally any recommendations).

A reader should be able to read your abstract and get a good understanding of what the report covers, without having to read the whole report.

1 Introduction

Your report must always start with an introduction that identifies and defines the topic, issue, or area of concern for the report.

1.1 What goes in it?

- Description of your area of investigation and theoretical perspective, which may include why the topic is important, interesting, problematic or relevant to you/the assignment.
- The scope and parameters of your topic – what does your report include and exclude?
- An explanation of the approach you intend to take and how you intend to accomplish this – the basis for your selection of the literature
- an outline of your topic

1.2 How long should it be?

Long enough so that the reader has a sufficient understanding of the topic to understand the rest of your report. At the same time it should not go into excessive detail. In general, probably between one and two pages.

1.3 When to write it?

It is generally best to create a draft statement at the beginning so that you know what you are planning to do. Then write the final version at the end, after writing the main body and discussion. This ensures that you do not have to constantly rewrite it as you write your main body text and also that the two fit together better.

Table 1: An example table.

Item	Quantity
Widgets	42
Gadgets	13

Table 2: A larger example table with centered.

Item	Quantity	third	Forth
Widgets	42	?	?
Gadgets	13	?	?
Thingymabobs	13	?	?

2 Some LaTeX Examples

LaTeX is a powerful type-setting and layouting software and covering all of its functionality is beyond the scope of this document. However, some examples of commonly used commands and features are listed below, to help you get started.

2.1 Sections

Use sectionYour Section Title and subsectionYour subsection title commands to organize your document. LaTeX handles all the formatting and numbering automatically.

If you place a labellabel-name after a section or subsection, then you can use reflabel-name to create a reference to it. This is often useful in the introduction (section 1) to refer to parts of the report.

2.2 Comments

Comments can be added to the margins of the document using the Here's a comment in the margin! todo command, as shown in the example on the right. You can also add inline comments too:

[inline, color=green!40]This is an inline comment.

2.3 Tables and Figures

LaTeX provides functionality for inserting both tables and images into your report.

2.3.1 Tables

Use the begintable and begintabular commands to create basic tables. See Table 1, for an example.

The table definition defines the actual table container. It is important to remember that LaTeX was design to create readable layouts. This means that while you can control what your table looks like, you cannot control where in the document the table is placed. That is why you must always use the captionTable caption command to give the table a brief summary that explains what it shows. After the caption command you can use the same labeltable-label command to give the table a unique label. As with any other label, you can then use reftable-label to add a reference from the text to the actual table, as is done above.

The table content is defined in the begintabular`{column definition}`. The column definition specifies how many columns your table has and how text in each column is aligned. Columns can be (l)eft aligned, (c)entered, (r)ight-aligned using the three letters `l`, `c`, or `r`. The number of these alignment letters in the column definition defines how many columns the table has (see Table 2). You can also create vertical separator lines by adding `|` characters.

Finally in the actual table definition itself, the individual columns in each row are separated using the `&` character. You must make sure that you define exactly as many columns in each row as you did in the column definition, otherwise LaTeX will throw an error. Each line must be ended by a double backslash \\, which is the LaTeX definition for a new line.

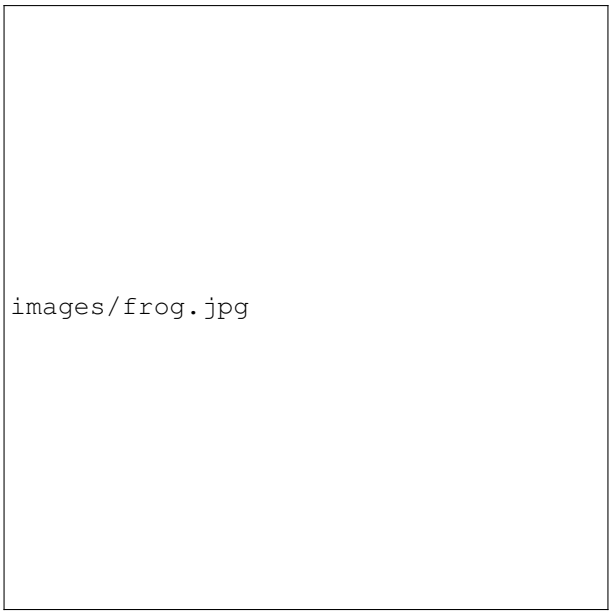


Figure 1: This is a figure caption.

Horizontal lines are defined using the toprule, midrule, and bottomrule commands. toprule should be used at the top of the table and bottomrule at the bottom. The midrule should be used for all other horizontal lines, although you should generally try to keep these to a minimum.

2.3.2 Images

LaTeX can handle JPEG, PNG, or PDF images. They are generally used in a similar way to tables, but are somewhat simpler. The image with its caption (see Figure 1) are defined using the beginfigure command. As with tables, the image will be placed by LaTeX and thus you should use the captionThe figure's caption command to include a brief description of the image's content. The labelfigure-label works as for tables.

The image itself is loaded using the `\includegraphics[options]{filename}` command. The filename is the path from where the `.tex` file is, to the location of the image file. The options parameter allows you to modify how the image is displayed. Common parameters are the `width` parameter, which lets you specify how wide the image should be and the `scale` parameter, which lets you scale the image up (values > 1) or down (values < 1).

2.4 Mathematics

LaTeX is great at typesetting mathematics. Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables with $\mathrm{E}[X_i] = \mu$ and $\mathrm{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n} = \frac{1}{n}\sum_{i}^{n} X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

2.5 Lists

LaTeX provides both enumerated lists using the beginenumerate command and bullet lists using beginitemize. Individual items in both types of lists are defined using the `\item` command.

1. Like this,
2. and like this.

- Like this,
- and like this.

3 Code inclusion

In general you should not include major amounts of code in your report, as the code is generally submitted separately. However, if you need to show some specific code that shows how you solved a very specific problem or overcame a coding issue, then you should include a code sample.

Code samples are included using the beginlstlisting[options] command. In general the `options` are used to define the language for syntax highlighting using `language=language_name`.

3.1 PHP

PHP code is highlighted by setting `language=php`

```
<?php
/* this is a php comment */
$username = $_POST["username"];
$password = $_POST["password"];

$pass = md5($password);

// another comment
if($username=="dave" and
   $pass=="5f4dcc3b5aa765d61d8327deb882cf99")
{
    echo "Hello␣David␣Walsh";
}
else
{
    echo "Login␣now";
}
?>
```

3.2 HTML, CSS, & JavaScript

Including mixed HTML, CSS, & JavaScript code samples is done by setting `style=htmlcssjs`.

```
<!DOCTYPE html>
<html>
  <head>
    <title>Listings Style Test</title>
    <meta charset="UTF-8">
    <style>
      /* CSS Test */
      * {
        padding: 0;
        border: 0;
        margin: 0;
      }
    </style>
    <link rel="stylesheet" href="css/style.css" />
  </head>
  <header> hey </header>
  <article> this is a article </article>
  <body>
    <!-- Paragraphs are fine -->
    <div id="box">
      <p>Hello World</p>
      <p>Hello World</p>
      <p id="test">Hello World</p>
      <p></p>
```

```
    </div>
    <div>Test</div>
    <!-- HTML script is not consistent -->
    <script src="js/benchmark.js"></script>
    <script>
      function createSquare(x, y) {
        // This is a comment.
        var square = document.createElement('div');
        square.style.width = square.style.height = '50px';
        square.style.backgroundColor = 'blue';

        /*
         * This is another comment.
         */
        square.style.position = 'absolute';
        square.style.left = x + 'px';
        square.style.top = y + 'px';

        var body = document.getElementsByTagName('body')[0];
        body.appendChild(square);
      };

      // Please take a look at +=
      window.addEventListener('mousedown', function(event) {
        var x = event.touches[0].pageX;
        var y = event.touches[0].pageY;
        var lookAtThis += 1;
      });
    </script>
  </body>
</html>
```

3.3 JavaScript

JavaScript code is highlighted by setting `language=javascript`

```
function hello(name) {
    alert('Hello ' + name);
}

hello('Mark & Dave');
```

3.4 Python

Python code is highlighted by setting `language=python`

```
def hello(name) {
    alert('Hello␣' + name);
}

hello('Mark␣&␣Dave');
```

4 Citations and References

LaTeX makes working with citations and references very easy. You define all your references in a file with the extension `.bib`. You can either create this bibtex file by hand or you can use an online tool like "Cite this For Me.com" ctfm$_2$015$and use the ir handy export tool$.

Each entry in the bibtex file has a unique identifier, which you can then use to reference that entry from your text. There are two ways to reference the entries. You can use the citetidentifier command if you want the reference to be included directly in the text. For example to indicate that baeza1999 wrote one of the definitive books in this area. Alternatively, you can use the citepidentifier command to place the reference in parenthesis. This is useful if you want to refer to an idea that is mentioned in a source, such as the concept of the information journey bates1989 or query analysis belkin2003.

You load your bibtex file by adding the
addbibresourcebibtext filename with extension command before the
begindocument (in what is known as the preamble). To output the bibliography, use
printbibliography anywhere in the main text. It is important to note here that only those entries in the bibtex file that you actually refer to in the main text are included in the bibliography.